



AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Project Domain

[Excel / Power BI with Copilot]

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:



Student Name

A.R. Anisha Priyatharshini



Student Email

anishapriyadharshini12@gmail.com



Guide

Uma Maheshwari

Institution:

Employee Payroll Data Cleaning and Financial Analysis

Dr. M.G.R. Janaki College Of Arts And Science For Women

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I express my heartfelt thanks to my project guide and mentors for their continuous guidance, encouragement, and valuable suggestions throughout the development of this project. Their support helped me successfully complete the project and improve my understanding of Excel-based analytics and AI-assisted reporting techniques.

I would also like to thank my institution and faculty members for providing the necessary resources, learning environment, and technical support required for the successful completion of this project.

Special thanks to Microsoft Excel, Microsoft Copilot, and ChatGPT for assisting in data analysis, visualization, and AI-generated insights used in this project.

Finally, I extend my gratitude to my family and friends for their constant encouragement, motivation, and support throughout the completion of this project report.

ABSTRACT OF THE PROJECT

The project titled “Employee Payroll Data Cleaning and Financial Analysis” focuses on transforming raw payroll data into meaningful financial insights using Microsoft Excel and Artificial Intelligence tools. Payroll systems are essential for managing employee salaries, bonuses, tax deductions, and payment records in organizations. However, payroll datasets often contain inconsistencies such as missing values, duplicate records, incorrect salary entries, invalid tax calculations, and inconsistent date formats, which affect the accuracy of financial reporting and decision-making.

The main objective of this project is to clean and preprocess payroll data to improve data accuracy and reliability. The project involves identifying and correcting errors, handling missing values, removing duplicate records, standardizing formats, and validating salary-related calculations using Excel tools and formulas.

After the cleaning process, financial analysis was performed using Excel formulas such as SUM, AVERAGE, IF, SUMIF, COUNTIF, and VLOOKUP along with Pivot Tables for department-wise salary analysis and payroll expense calculation. Various charts and dashboards were created to visualize employee salary distribution, bonus analysis, tax deductions, and department-wise payroll expenses.

Additionally, Artificial Intelligence tools such as ChatGPT and Microsoft Copilot were integrated into the project to generate automated summaries, payroll insights, and recommendations. AI-generated narratives helped simplify data interpretation and improve reporting efficiency.

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CHAPTER 1

Introduction

1.1 Problem Statement

Organizations generate large amounts of payroll data related to employee salaries, bonuses, tax deductions, and payment records. However, payroll datasets often contain inconsistencies such as missing values, duplicate records, incorrect salary calculations, invalid tax entries, and inconsistent date formats. These issues reduce data accuracy and make payroll management difficult.

Managing payroll data manually becomes time-consuming and increases the possibility of human errors, especially when organizations handle large employee datasets. Incorrect payroll records may lead to salary miscalculations, financial discrepancies, and poor decision-making.

This project aims to address these challenges by:

- Cleaning and preprocessing employee payroll data
- Performing financial analysis using Excel formulas and Pivot Tables
- Creating charts and dashboards for visualization
- Integrating AI tools to generate automated insights and summaries

The main goal of the project is to transform raw payroll data into a clean, structured, and meaningful format that improves payroll accuracy and supports better financial decision-making.

1.2 Motivation

The main motivation behind this project is to improve payroll data accuracy and simplify financial analysis using Excel and Artificial Intelligence tools.

The project also helps in understanding how AI tools such as ChatGPT and Microsoft Copilot can generate insights and summaries from payroll data efficiently.

1.3 Objectives

- To collect and analyze a payroll dataset
- To remove duplicate records and missing values
- To correct incorrect salary and tax entries
- To perform financial analysis using Excel formulas and Pivot Tables

1.4 Scope of the Project

This project focuses on employee payroll data cleaning and financial analysis using Microsoft Excel and AI tools. The project includes cleaning payroll data, analysing salary information, creating visual reports, and generating AI-assisted insights.

Limitations (Out of Scope):-

- The project uses a static dataset
- Real-time payroll integration is not included
- Advanced machine learning models are not implemented

CHAPTER 2

Literature Survey

A literature survey helps in understanding existing tools, techniques, and technologies related to the project. In the field of data analytics, Microsoft Excel and Business Intelligence tools are widely used for data cleaning, analysis, and visualization.

Many organizations use payroll systems to manage employee salary records, tax deductions, and financial reporting. However, payroll datasets often contain errors and inconsistencies that affect data accuracy. Data cleaning techniques help in improving the quality of payroll information and making analysis more reliable.

Artificial Intelligence tools such as ChatGPT and Microsoft Copilot are now being used to generate automated insights, summaries, and recommendations from datasets. These tools reduce manual effort and improve reporting efficiency.

2.1 Existing Work and Research

Previous studies and business practices mainly focus on payroll management systems, salary processing, financial reporting, and employee compensation analysis using Excel and Business Intelligence tools. Many organizations use payroll software and spreadsheet-based systems to maintain employee salary records and calculate payroll expenses.

Existing systems commonly use formulas, Pivot Tables, and charts for analysing salary distribution, tax deductions, and department-wise payroll expenses. Dashboards are also used to present payroll information visually for easier understanding and reporting.

Although many payroll systems provide reporting and visualization features, AI-based insight generation and automated explanations are still limited in most organizations. Many systems still depend on manual interpretation of payroll data.

2.2 Tools and Technologies Review

The following tools and technologies are used in this project:

Microsoft Excel

Microsoft Excel is used for data cleaning, salary calculations, Pivot Tables, formulas, and financial analysis. It helps in removing duplicate records, correcting errors, handling missing values, and analysing payroll data efficiently.

Microsoft Power BI

Power BI is used for creating dashboards, charts, and visual reports. It helps in visualizing department-wise salary distribution, payroll expenses, and employee compensation trends.

ChatGPT / Microsoft Copilot

AI tools such as ChatGPT and Microsoft Copilot are used to generate summaries, payroll insights, and recommendations from the analysed data. These tools improve reporting efficiency and simplify data interpretation.

2.3 Gaps Identified

- Manual payroll analysis requires more time and effort
- Existing payroll systems provide limited automated insights
- Large payroll datasets are difficult to manage manually
- Many systems lack AI integration for intelligent reporting and analysis
- Manual verification increases the chances of payroll errors and inconsistencies

CHAPTER 3

Proposed Methodology

3.1 System Design

The proposed system is designed to clean and analyse employee payroll data using Microsoft Excel and Artificial Intelligence tools. The project follows a structured process from data collection to final analysis and visualization.

The workflow of the project includes the following steps:

- **Data Collection** : Collecting the payroll dataset containing employee salary details, bonuses, tax deductions, and payment records.
- **Data Cleaning** : Identifying and correcting errors such as missing values, duplicate records, incorrect salary entries, and inconsistent date formats.
- **Financial Analysis** : Performing salary analysis using Excel formulas, Pivot Tables, and calculations.
- **Visualization** : Creating charts and dashboards to represent payroll insights visually.
- **AI Integration** : Using AI tools such as ChatGPT and Microsoft Copilot to generate summaries, insights, and recommendations.

The system helps transform raw payroll data into a clean and structured format that improves reporting and decision-making.

3.2 Modules / Components Used

The project is divided into the following modules:

- Data Collection Module : Used for importing and understanding the payroll dataset containing employee salary information.
- Data Cleaning Module : Used for removing duplicates, handling missing values, correcting invalid entries, and standardizing data formats.
- Financial Analysis Module : Used for salary calculations, tax analysis, bonus analysis, and department-wise payroll analysis using Excel formulas and Pivot Tables.
- Visualization Module : Used for creating charts, graphs, and dashboards to represent payroll information visually.
- AI Integration Module : Used for generating AI-based summaries, payroll insights, and recommendations using ChatGPT and Microsoft Copilot.

3.3 Data Flow Diagram (DFD)

Represents the overall system flow:-

PAYROLL DATASET → DATA CLEANING → FINANCIAL ANALYSIS → DASHBOARD & AI INSIGHTS

DFD Level 0 — Context Diagram:-

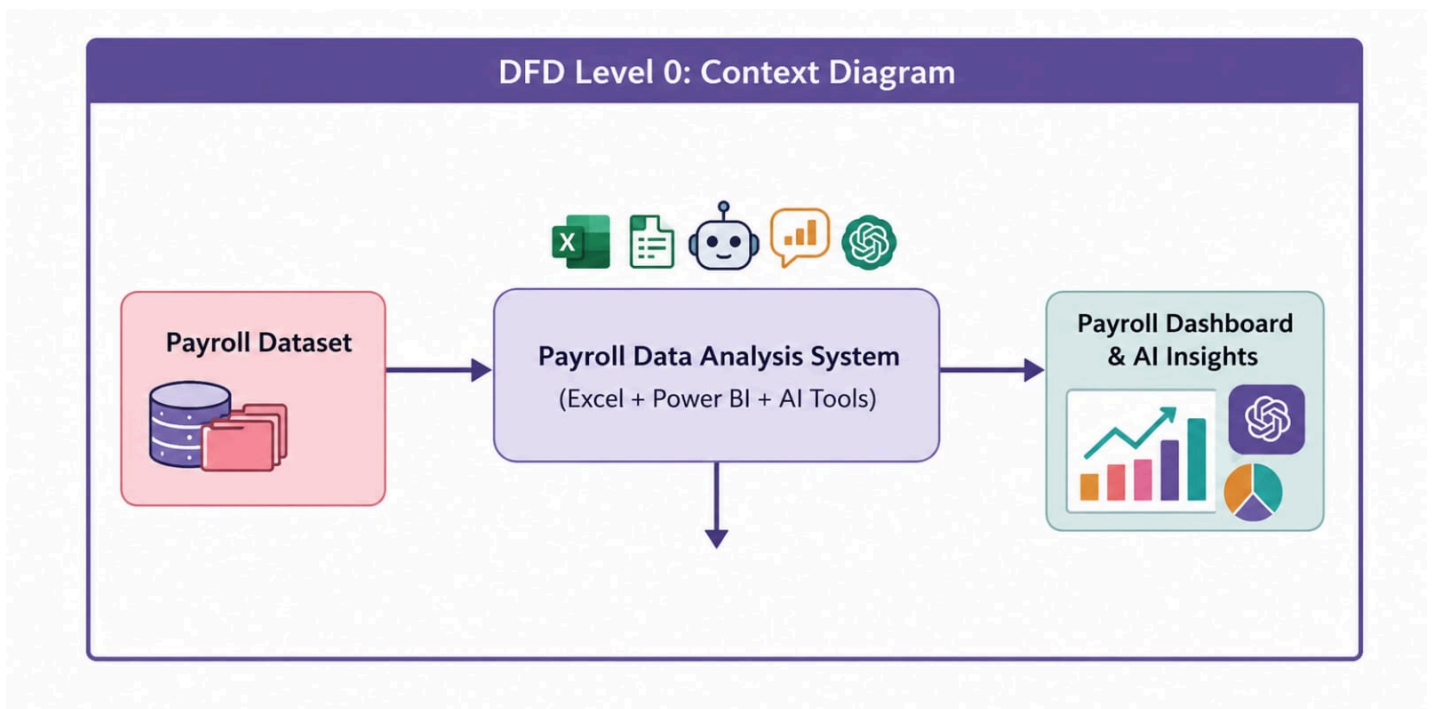


Figure : 1 DFD Employee Payroll Flow

The system receives payroll data as input, processes the data through cleaning and analysis techniques, and produces dashboards and AI-generated insights as output.

DFD Level 1 — Detailed Process Flow

Breakdown of processes:

1. Data Collection
2. Data Cleaning and Validation
3. Salary and Tax Analysis
4. Dashboard and Chart Creation
5. AI-Based Insight Generation

3.4 Advantages

- Reduces manual effort in payroll analysis
- Improves payroll data accuracy and consistency
- Helps identify salary and tax-related insights easily
- Provides visual dashboards for better understanding
- Integrates AI tools for automated reporting and summaries
- Improves decision-making and reporting efficiency

3.5 Requirement Specification Hardware Requirements

Component	Specification
Processor	Intel Corei3orhigher (i5/i7 recommended)
RAM	Minimum4GB(8GB recommended)
Storage	Minimum50GBfree disk space
Display	1366 x 768resolution or higher
Internet	RequiredforCopilot / Chatbot access

3.6 Software Requirements

Software	Details
OperatingSystem	Windows 10 / 11 or macOS
MicrosoftExcel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BIDesktop	Latest version (free download from Microsoft)
MicrosoftCopilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
WebBrowser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

4.1 Data Collection and Dataset Description

The project uses an employee payroll dataset containing employee details such as employee ID, department, salary, bonus, tax deduction, and payment information. The dataset was collected and maintained in Microsoft Excel for analysis and reporting purposes.

	A	B	C	D	E	F	G	H	I	J	K
1	Employee_ID	Employee_Name	Department	Designation	Date_of_Joining	Basic_Salary	Allowances	Bonus	Tax_Deduction	Net_Salary	Payment_Date
2	EMP001	Rahul Sharma	HR	HR Executive	15-01-2020	45000	5000	4000	4500	49,500	31-05-2024
3	EMP002	Priya Verma	Finance	Accountant	20/02/2019	52,000	6000	5000	5200	57,800	31/05/2024
4	EMP002	Priya Verma	Finance	Accountant	20-02-2019	52000	6000	5000	5200	57800	31-05-2024
5	EMP003	Amit Kumar	IT	Software Engineer	10-03-21	60,000	7000	6000	6000	67000	31-05-2024
6	EMP004	Anjali Singh	Marketing	Marketing Executive	05-07-2020	47000	5500	4000	4700	50,800	31/05/2024
7	EMP005	Vikram Patel	Sales	Sales Executive	12/11/2018	48,000.00	5000	4500	4800	51,900.00	31-05-2024
8	EMP006	Neha Gupta	HR	HR Executive	18-06-2019	46000	5000	4000	4600	50400	31/05/2024
9	EMP007	Saurabh Yadav	IT	System Analyst	22-08-2021	62000	7500	6000	6200	69,300	31-05-2024
10	EMP008	Pooja Mehta	Finance	Finance Executive	30/09/2018	51000	6000	5000	5100	57900	31-05-2024
11	EMP009	Rohit Das	IT	Software Engineer	14-04-2020	58,000	7000	5500	5800	64,700	31/05/2024
12	EMP010	Kavita Joshi	Marketing	Marketing Executive	25/01/2019	48000	5,500	4500	4800	52400	31-05-2024
13	EMP011	Deepak Sharma	Sales	Sales Executive	11-05-2021	49000	5000	4500	4900	52,600	31-05-2024
14	EMP012	Sneha Reddy	HR	HR Executive	19/12/2018	45,000	5000	4000	4500	49500	31/05/2024
15	EMP013	Arjun Nair	IT	Software Engineer	07-03-2020	61000	7500	6000	6100	68,400	31-05-2024
16	EMP014	Meera Iyer	Finance	Accountant	29-07-2019	50,000	6000	5000	5000	57,000	31-05-2024
17	EMP015	Manish Kumar	Sales	Sales Executive	03/09/2020	47000	5000	4500	4700	51800	31/05/2024
18	EMP016	Riya Malhotra	Marketing	Marketing Executive	21-02-2021	49,000	5500	4500	4900	53200	31-05-2024
18	EMP017	Mohit Verma	IT	System Analyst	16-10-2019	60000	7000	6000	6000	67000	31-05-2024
20	EMP018	Divya Jain	Finance	Finance Executive	04/06/2020	52000	6000	5000	5200	57800	31/05/2024
21	EMP019	Kunal Singh	Sales	Sales Executive	28-11-2018	48,000	5000	4500	4800	51,900	31-05-2024
22	EMP020	Singh Kaur	HR	HR Executive	08-01-2021	46000	5000	4000	4600	50400	31-05-2024

Figure : 2 Employee Payroll Dataset (Uncleaned Version)

In the Figure : 2, The dataset initially contained inconsistencies such as duplicate records, missing values, incorrect salary entries, and inconsistent formatting. These issues affected the accuracy of payroll analysis and required proper data cleaning before further processing.

4.2 Data Cleaning Process

Data cleaning was performed using Microsoft Excel to improve the quality and accuracy of the payroll dataset. The cleaning process included identifying duplicate records, correcting missing values, standardizing date formats, and verifying salary and tax calculations.

	A	B	C	D	E	F	G	H	I	J	K
1	Employee_ID	Employee_Name	Department	Designation	Date_of_Joining	Basic_Salary	Allowances	Bonus	Tax_Deduction	Net_Salary	Payment_Date
2	EMP001	Rahul Sharma	HR	HR Executive	15-01-2020	45,000.00	5,000.00	4,000.00	4,500.00	49,500.00	31-05-2024
3	EMP002	Priya Verma	Finance	Accountant	20-02-2019	52,000.00	6,000.00	5,000.00	5,200.00	57,800.00	31-05-2024
4	EMP003	Amit Kumar	IT	Software Engineer	10-03-2021	60,000.00	7,000.00	6,000.00	6,000.00	67,000.00	31-05-2024
5	EMP004	Anjali Singh	Marketing	Marketing Executive	05-07-2020	47,000.00	5,500.00	4,000.00	4,700.00	50,800.00	31-05-2024
6	EMP005	Vikram Patel	Sales	Sales Executive	12-11-2018	48,000.00	5,000.00	4,500.00	4,800.00	51,900.00	31-05-2024
7	EMP006	Neha Gupta	HR	HR Executive	18-06-2019	46,000.00	5,000.00	4,000.00	4,600.00	50,400.00	31-05-2024
8	EMP007	Saurabh Yadav	IT	System Analyst	22-08-2021	62,000.00	7,500.00	6,000.00	6,200.00	69,300.00	31-05-2024
9	EMP008	Pooja Mehta	Finance	Finance Executive	30-09-2018	51,000.00	6,000.00	5,000.00	5,100.00	57,900.00	31-05-2024
10	EMP009	Rohit Das	IT	Software Engineer	14-04-2020	58,000.00	7,000.00	5,500.00	5,800.00	64,700.00	31-05-2024
11	EMP010	Kavita Joshi	Marketing	Marketing Executive	25-01-2019	48,000.00	5,500.00	4,500.00	4,800.00	52,400.00	31-05-2024
12	EMP011	Deepak Sharma	Sales	Sales Executive	11-05-2021	49,000.00	5,000.00	4,500.00	4,900.00	52,600.00	31-05-2024
13	EMP012	Sneha Reddy	HR	HR Executive	19-12-2018	45,000.00	5,000.00	4,000.00	4,500.00	49,500.00	31-05-2024
14	EMP013	Arjun Nair	IT	Software Engineer	07-03-2020	61,000.00	7,500.00	6,000.00	6,100.00	68,400.00	31-05-2024
15	EMP014	Meera Iyer	Finance	Accountant	29-07-2019	50,000.00	6,000.00	5,000.00	5,000.00	57,000.00	31-05-2024
16	EMP015	Manish Kumar	Sales	Sales Executive	03-09-2020	47,000.00	5,000.00	4,500.00	4,700.00	51,800.00	31-05-2024
17	EMP016	Riya Malhotra	Marketing	Marketing Executive	21-02-2021	49,000.00	5,500.00	4,500.00	4,900.00	53,200.00	31-05-2024
18	EMP017	Mohit Verma	IT	System Analyst	16-10-2019	60,000.00	7,000.00	6,000.00	6,000.00	67,000.00	31-05-2024
19	EMP018	Divya Jain	Finance	Finance Executive	04-06-2020	52,000.00	6,000.00	5,000.00	5,200.00	57,800.00	31-05-2024
20	EMP019	Kunal Singh	Sales	Sales Executive	28-11-2018	48,000.00	5,000.00	4,500.00	4,800.00	51,900.00	31-05-2024
21	EMP020	Simran Kaur	HR	HR Executive	09-01-2021	46,000.00	5,000.00	4,000.00	4,600.00	50,400.00	31-05-2024

Figure : 3 Cleaned Version Of Dataset

In Figure : 3, Excel features such as sorting, filtering, conditional formatting, and formulas were used to detect and correct errors in the dataset. After cleaning, the payroll data became more organized, accurate, and ready for analysis.

4.3 Financial Analysis

Financial analysis was performed using Excel formulas and Pivot Tables to generate useful payroll insights. Salary distribution, bonus allocation, department-wise payroll expenses, and tax deductions were analysed using different calculations and summary reports.

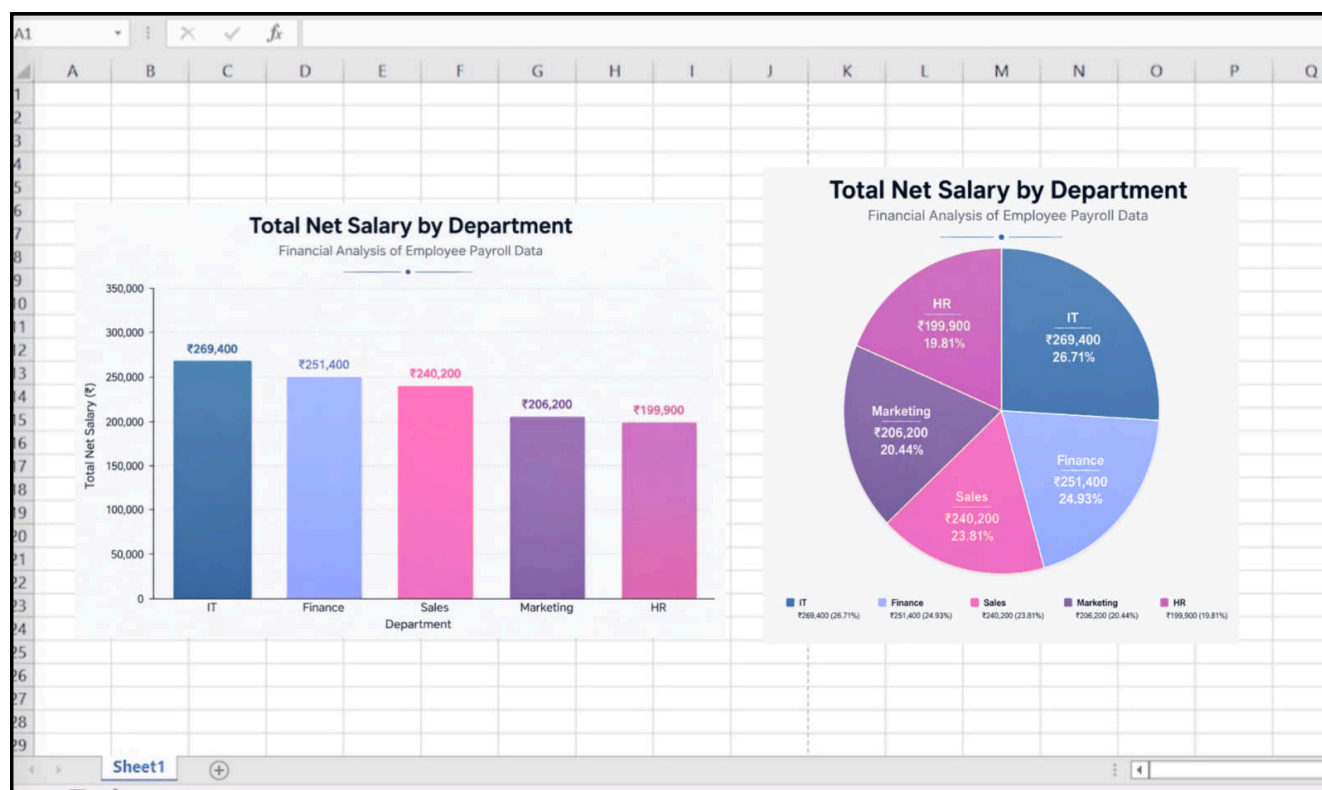


Figure : 4 Total Nat Salary Of Department

In Figure : 4, Formulas such as SUM, AVERAGE, COUNTIF, and IF statements were used to perform payroll-related calculations. The analysis helped in understanding employee compensation patterns and overall payroll expenses more effectively in a visual bar chart and pie chart format.

4.4 Visualization and Dashboard Results

Charts, graphs, and dashboards were created to visually represent payroll data and financial insights. Bar charts, pie charts, and Pivot Charts were used to display salary distribution, department-wise payroll expenses, and bonus analysis.

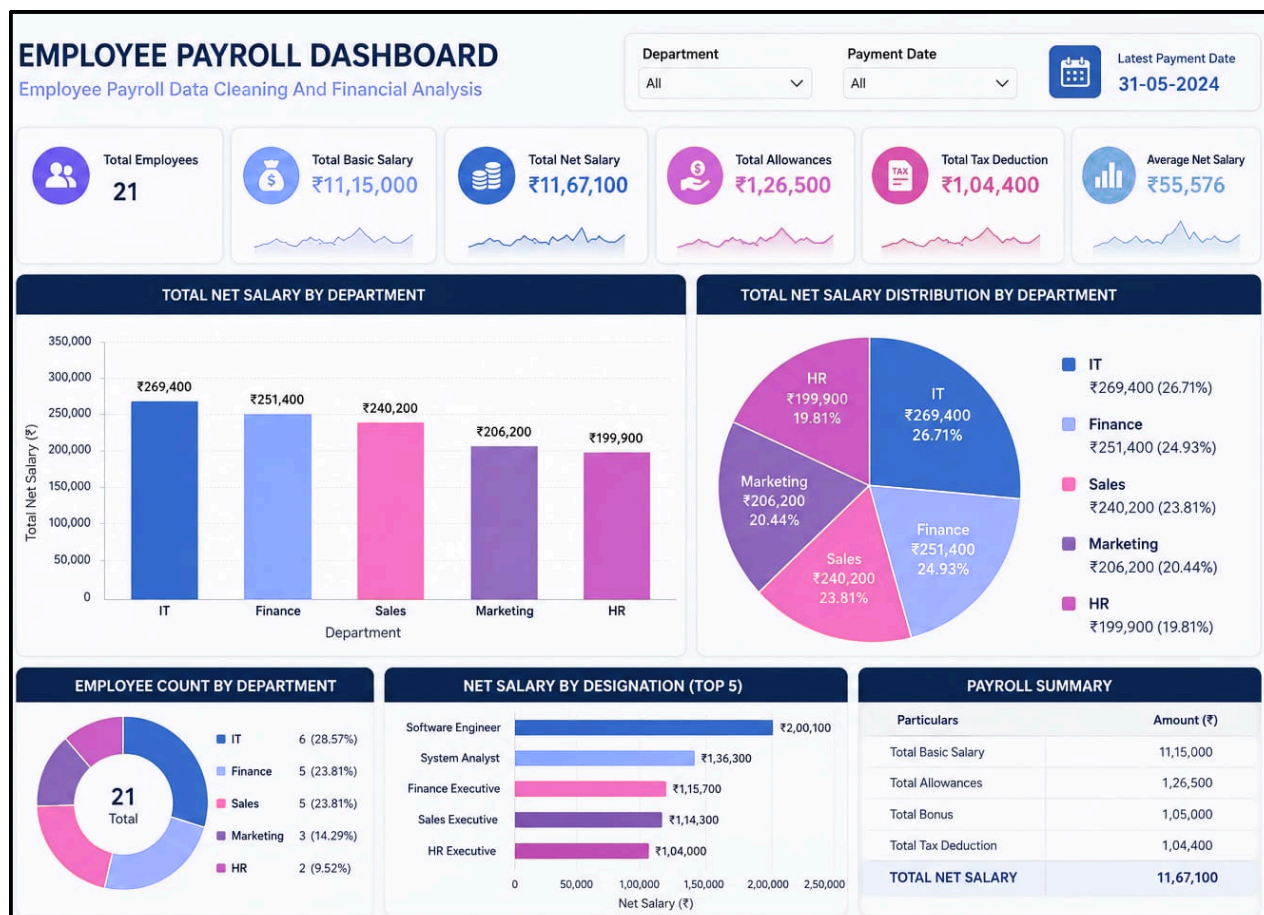


Figure : 5 Employee Payroll Dashboard

In Figure : 5, The dashboard provided a clear visual summary of payroll information and helped in understanding trends and comparisons more easily. Visualization improved the readability and presentation of payroll data.

4.5 AI Tool Integration Results

AI tools such as ChatGPT and Microsoft Copilot were used to enhance analysis.

Tool Used:-

Open AI ChatGPT / Microsoft Copilot

Sample Prompts:-

- “Summarize department-wise payroll expenses”
- “Identify trends in employee salary and tax deductions”
- “Provide recommendations based on payroll analysis”

AI Output:-

- Generated summaries of payroll trends
- Identified departments with highest salary expenses
- Suggested improvements for payroll accuracy and financial management

Impact of AI Integration:-

- Reduced manual effort in analysis
- Faster report generation
- Improved understanding of data through natural language insights

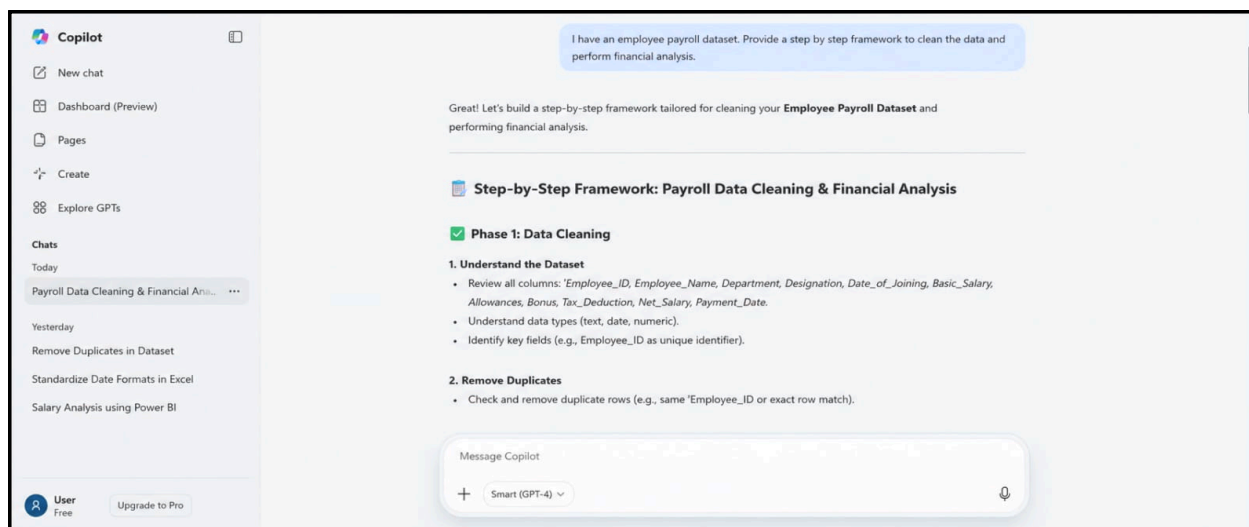


Figure : 6 Microsoft Copilot Integration with the project

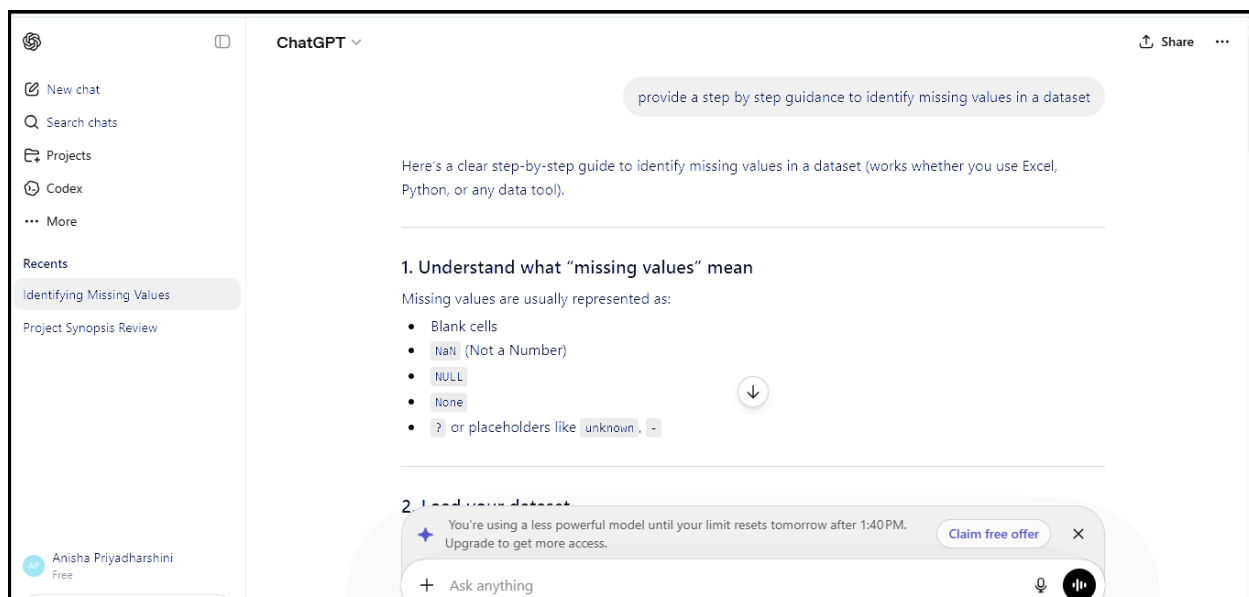


Figure : 7 Open AI ChatGPT Integration with the project

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

- Payroll data cleaning improved data accuracy and consistency
- Duplicate records and missing values were corrected successfully
- Financial analysis helped identify salary and tax patterns
- Department-wise payroll expenses were analysed effectively
- Charts and dashboards improved data visualization and understanding
- AI tools generated useful payroll summaries and insights
- AI integration reduced manual effort in reporting and analysis

5.2 Limitations

Although the project achieved its objectives, some limitations exist:

- The project uses a static payroll dataset
- Real-time payroll updates are not included
- Advanced machine learning models are not implemented
- The analysis is limited to Excel and AI-based tools

5.3 Project Link

- GitHub Link of the Project [<https://github.com/anishapriyadharshini/Employee-Payroll-Data-Cleaning-And-Financial-Analysis-.git>]
- Kaggle Dataset <https://www.kaggle.com>

5.4 Future Work

The project can be further enhanced in the following ways:

- Real-time payroll systems can be integrated
- Cloud database support can be added
- Advanced AI and machine learning models can be implemented
- Fraud detection and payroll prediction features can be added
- Interactive web-based dashboards can be developed

5.5 Conclusion

The project demonstrates the importance of payroll data cleaning and financial analysis in improving payroll accuracy and reporting efficiency. Microsoft Excel and AI tools were effectively used to clean payroll data, perform analysis, create visual dashboards, and generate meaningful insights.

The project helped in understanding how data analytics and Artificial Intelligence can support better payroll management and financial decision-making in organizations.

REFERENCES

List of all references, research papers, tutorials, and documentation used in project format below:-

1. **Microsoft Corporation**

Microsoft Excel Documentation <https://support.microsoft.com/excel>

2. **Microsoft Corporation**

Power BI Documentation <https://learn.microsoft.com/power-bi>

3. **Microsoft Corporation**

Microsoft Copilot – AI Assistant <https://copilot.microsoft.com>

4. **Edunet Foundation**

AICWFellowshipProgramGuidelines, 2026

5. **Kaggle Inc.**

Employee Payroll Dataset <https://www.kaggle.com>

6. **Tutorials and learning resources from:**

- YouTube (Power BI & Excel tutorials)
 - GitHub repositories on data analytics projects
-

APPENDICES

Appendix A- Sample Dataset

	A	B	C	D	E	F	G	H	I	J	K
1	Employee_ID	Employee_Name	Department	Designation	Date_of_Joining	Basic_Salary	Allowances	Bonus	Tax_Deduction	Net_Salary	Payment_Date
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18	EMP017	Mohit Verma	IT	System Analyst	16-10-2019	60,000.00	7,000.00	6,000.00	6,000.00	67,000.00	31-05-2024
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20	EMP019	Kunal Singh	Sales	Sales Executive	28-11-2018	48,000.00	5,000.00	4,500.00	4,800.00	51,900.00	31-05-2024
21	EMP020	Simran Kaur	HR	HR Executive	09-01-2021	46,000.00	5,000.00	4,000.00	4,600.00	50,400.00	31-05-2024

Figure : 8 Sample Dataset of Employee Payroll (Cleaned Version)

In Figure 8, I presented the cleaned Employee Payroll Dataset used for payroll data cleaning and financial analysis.

The dataset contains employee details such as Employee ID, Department, Designation, Salary Components, Tax Deduction, Net Salary, and Payment Date in a structured format. After preprocessing and standardization, the dataset became suitable for financial analysis, visualization, and dashboard creation.

Appendix B- Key Excel Formulas / DAX Measures Used

Excel Formulas Used:

- SUM Formula – Used to calculate total salary, bonus, and payroll expenses
- AVERAGE Formula – Used to calculate average employee salary
- IF Formula – Used to apply conditions for salary and tax calculations
- COUNTIF Formula – Used to count duplicate or specific records in the dataset
- VLOOKUP Formula – Used to retrieve employee details from the dataset
- Pivot Tables – Used to summarize and analyse payroll data efficiently

Appendix C — AI Prompts Used

Some of the key prompts used in AI tools:

- “Summarize department-wise payroll expenses”
- “Identify salary and tax trends”
- “Generate payroll analysis summary”
- “Provide recommendations based on payroll data”

Appendix D — Screenshot Gallery

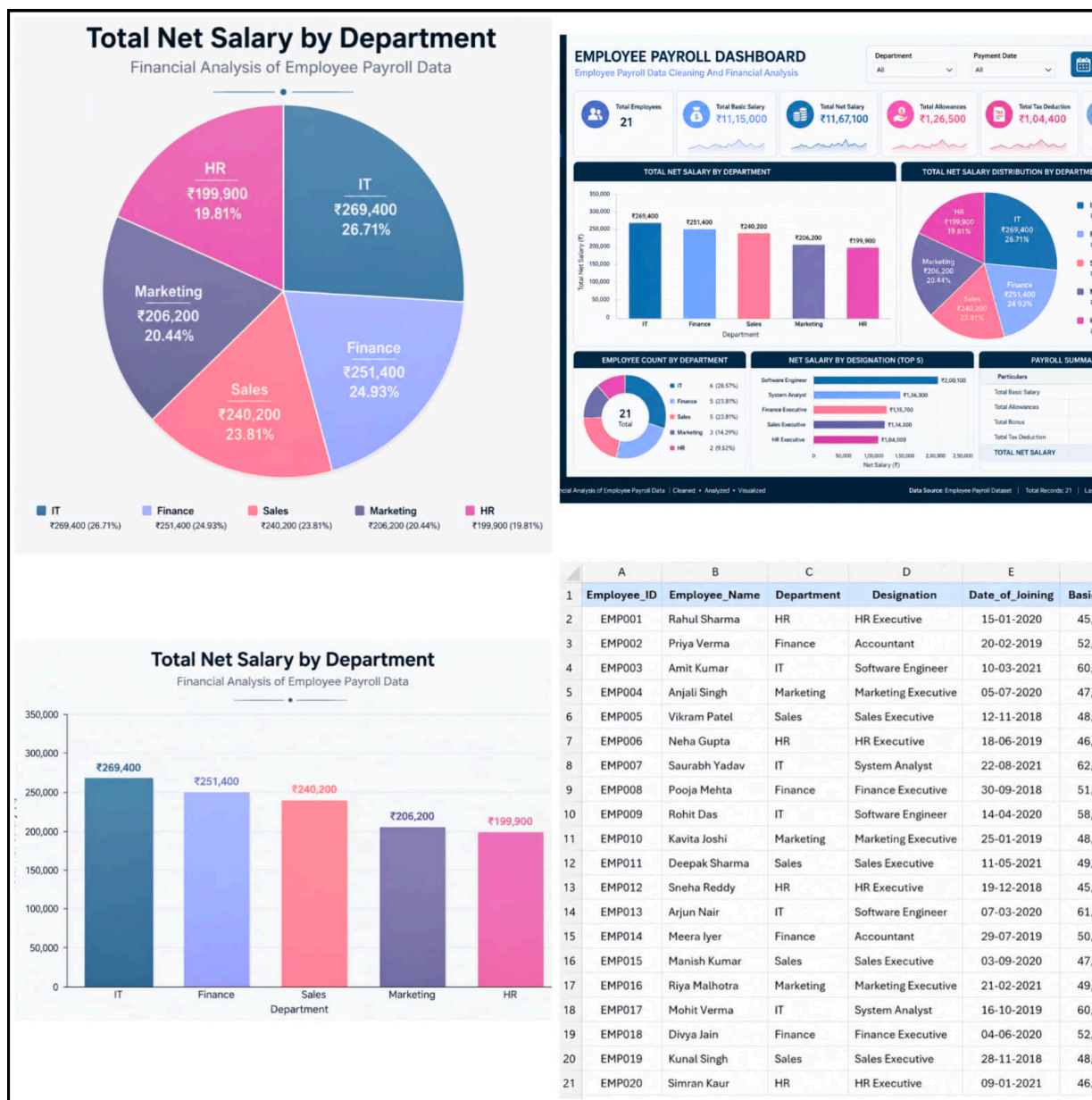


Figure : 9 Screenshots Of Employee Payroll Dashboard And Cleaned Dataset

Thank You